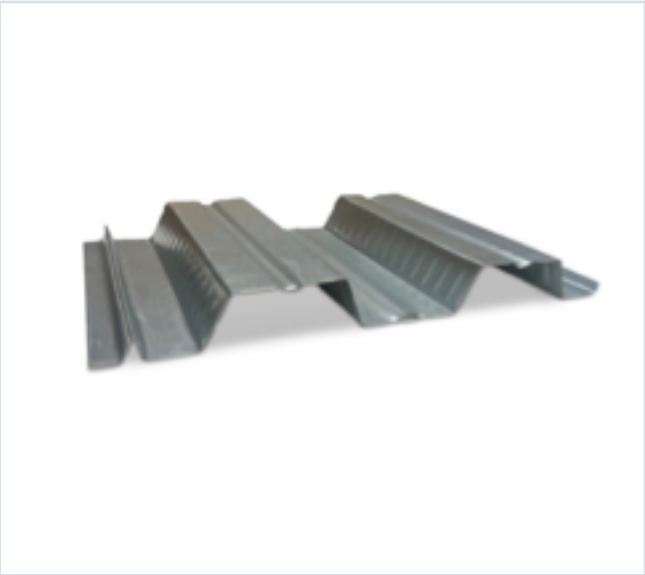


DECKING PRODUCT SUBMITTAL

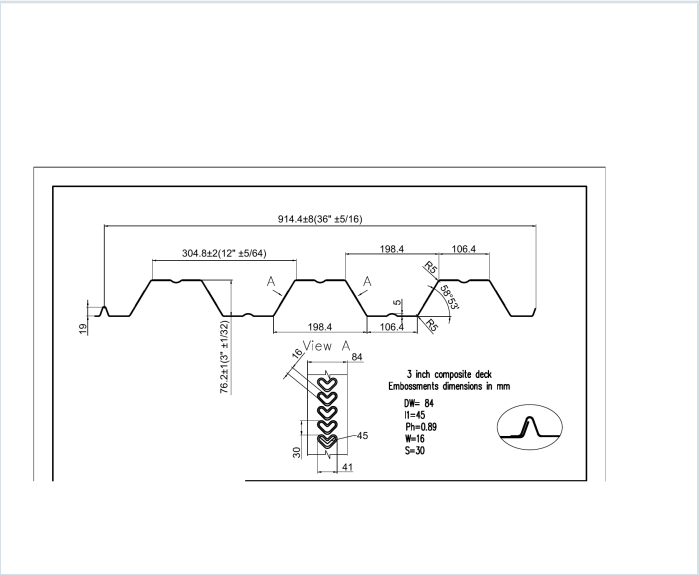
3CD-FY50-20-G40

PRODUCT	GRADE	GAUGE	COATING
3" Composite Deck	Grade 50	20 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 20 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0358	2.1	50	0.525	0.559	15709	16747	0.908	0.870

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 62, 63, 64, 65, 66
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

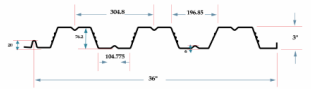
3CD-FY50-20-G40 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 62

GRADE 50 STEEL



Section Properties									
Gage	Design Thickness (inches)	Weight (lb/ft)	E _x (ksi)	S _x (in ³ /ft)	S _y (in ³ /ft)	Se (in ⁴ /ft)	ASD (D + 16T) M _y /O (in ⁴ /ft)	M _y /O (in ⁴ /ft)	I _y (in ⁴ /ft)
22	0.0295	17	50	0.397	0.426	10709	10747	0.908	0.970
20	0.0358	21	50	0.525	0.559	13922	13962	1.274	1.360
18	0.0474	27	50	0.796	0.796	23022	23062	1.694	1.800
16	0.0598	35	50	1.009	1.005	30000	30030	2.244	2.380

Shear and Web Crippling									
Gage	V _y /O (lb/ft)	Web Crippling (R _y), lb/ft One Flange Loading				Web Crippling (R _y), lb/ft Interior Bearing			
		2"	4"	6"	8"	2"	4"	6"	8"
22	789	403	463	575	643	728	799	999	1143
20	2385	578	663	754	858	1045	1143	1413	1543
18	4272	874	1010	1224	1375	1700	1877	2244	2443
16	7370	1497	1696	1864	2135	2707	2936	3544	3843

Allowable Uniform Downward Loads, ASD (PSF)									
Span	Gage	6"	7"	8"	9"	10"	12"	14"	16"
Single	22	207	214	224	234	244	254	264	274
	20	291	304	314	324	334	344	354	364
	18	441	454	464	474	484	494	504	514
	16	599	614	624	634	644	654	664	674
Double	22	236	244	254	264	274	284	294	304
	20	340	354	364	374	384	394	404	414
	18	441	454	464	474	484	494	504	514
	16	599	614	624	634	644	654	664	674
Triple	22	226	234	244	254	264	274	284	294
	20	330	344	354	364	374	384	394	404
	18	431	444	454	464	474	484	494	504
	16	589	604	614	624	634	644	654	664

Notes:
1. All section properties and ASD (D + 16T) uniform loads are calculated in accordance with AISI S100-2017, AISI S100-2012 and AISI S100-2008.
2. Loads shown in tables are uniformly distributed downward loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased by increasing clear span dimensions.
3. Bending Moment formula used for forward stress limitations are: Simple and Two Span: $M = \frac{wL^2}{8}$; Three Span: $M = \frac{wL^2}{16}$.
4. Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 63

span	Gage	6"	7"	8"	9"	10"	12"	14"	16"
Single	22	207	214	224	234	244	254	264	274
	20	291	304	314	324	334	344	354	364
	18	441	454	464	474	484	494	504	514
	16	599	614	624	634	644	654	664	674
Double	22	236	244	254	264	274	284	294	304
	20	340	354	364	374	384	394	404	414
	18	441	454	464	474	484	494	504	514
	16	599	614	624	634	644	654	664	674
Triple	22	226	234	244	254	264	274	284	294
	20	330	344	354	364	374	384	394	404
	18	431	444	454	464	474	484	494	504
	16	589	604	614	624	634	644	654	664

Note:
For loads that cause L/250 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/160 Deflection, multiply by 0.67.

Construction Span Table – 20 psf Construction Load									
Normal Weight Concrete (150 pcf)					Lightweight Concrete (85 pcf)				
Total Slab Depth	Deck Type	Maximum Unfinished Clear Span	1 span	2 span	3 span	Total Slab Depth	Deck Type	Maximum Unfinished Clear Span	1 span
6.5 (F+200) 52 PSF	3x12x22 ga	10' 3"	11' 4"	11' 9"	11' 9"	6.5 (F+200) 52 PSF	3x12x22 ga	10' 3"	11' 4"
	3x12x20 ga	10' 3"	11' 4"	11' 9"	11' 9"		3x12x20 ga	10' 3"	11' 4"
	3x12x18 ga	10' 3"	11' 4"	11' 9"	11' 9"		3x12x18 ga	10' 3"	11' 4"
	3x12x16 ga	10' 3"	11' 4"	11' 9"	11' 9"		3x12x16 ga	10' 3"	11' 4"
6.5 (F+300) 58 PSF	3x12x22 ga	9' 9"	10' 10"	10' 11"	10' 11"	6.5 (F+300) 58 PSF	3x12x22 ga	9' 9"	10' 10"
	3x12x20 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x20 ga	9' 9"	10' 10"
	3x12x18 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x18 ga	9' 9"	10' 10"
	3x12x16 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x16 ga	9' 9"	10' 10"
7.5 (F+400) 70 PSF	3x12x22 ga	9' 9"	10' 10"	10' 11"	10' 11"	7.5 (F+400) 70 PSF	3x12x22 ga	9' 9"	10' 10"
	3x12x20 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x20 ga	9' 9"	10' 10"
	3x12x18 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x18 ga	9' 9"	10' 10"
	3x12x16 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x16 ga	9' 9"	10' 10"
7.5 (F+500) 78 PSF	3x12x22 ga	9' 9"	10' 10"	10' 11"	10' 11"	7.5 (F+500) 78 PSF	3x12x22 ga	9' 9"	10' 10"
	3x12x20 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x20 ga	9' 9"	10' 10"
	3x12x18 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x18 ga	9' 9"	10' 10"
	3x12x16 ga	9' 9"	10' 10"	10' 11"	10' 11"		3x12x16 ga	9' 9"	10' 10"

Note:
Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

3CD-FY50-20-G40 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 64

22 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0	8'-6	9'-0	9'-6	10'-0	10'-6	11'-0
5	46	272	258	240	196	165	147	132
5.5	52	329	288	254	225	200	179	160
6	58	388	340	300	266	237	212	190
6.5	64	400	354	317	308	274	248	223
7	70	400	400	386	351	313	280	252
7.5	76	400	400	400	395	352	316	284

Slab Thickness (Inches)	11'-6	12'-0	12'-6	13'-0	13'-6	14'-0	14'-6	15'-0
5	118	127	16	87	79	11	65	59
5.5	144	150	117	106	96	88	80	72
6	171	154	140	127	115	105	95	87
6.5	198	175	162	147	134	122	111	102
7	227	205	186	169	154	140	128	117
7.5	256	235	210	191	174	158	145	132

20 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0	8'-6	9'-0	9'-6	10'-0	10'-6	11'-0
5	46	329	288	254	225	200	180	162
5.5	52	397	349	308	273	244	218	196
6	58	460	400	363	323	288	258	232
6.5	64	400	400	400	374	334	299	270
7	70	400	400	400	400	381	342	308
7.5	76	400	400	400	400	400	385	347

Slab Thickness (Inches)	11'-6	12'-0	12'-6	13'-0	13'-6	14'-0	14'-6	15'-0
5	146	132	115	108	96	80	82	75
5.5	177	160	145	132	120	100	100	92
6	210	180	172	157	143	121	120	110
6.5	244	221	201	185	167	140	140	130
7	278	252	229	209	189	175	160	147
7.5	314	285	259	236	216	198	181	166

18 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0	8'-6	9'-0	9'-6	10'-0	10'-6	11'-0
5	46	400	400	381	339	303	273	246
5.5	52	400	400	400	397	355	320	289
6	58	400	400	400	400	400	370	334
6.5	64	400	400	400	400	400	400	368
7	70	400	400	400	400	400	400	400
7.5	76	400	400	400	400	400	400	400

Slab Thickness (Inches)	11'-6	12'-0	12'-6	13'-0	13'-6	14'-0	14'-6	15'-0
5	223	203	185	169	154	142	130	120
5.5	262	238	217	198	182	162	148	136
6	303	275	251	230	210	193	178	164
6.5	343	314	287	262	240	221	204	188
7	384	354	324	296	272	250	230	213
7.5	400	397	362	331	304	280	258	238

STEEL CATALOG PAGE 66

Slab Thickness (Inches)	Weight (psf)	8'-0	8'-6	9'-0	9'-6	10'-0	10'-6	11'-0
5	37	400	400	358	328	294	265	240
5.5	42	400	400	400	385	345	313	281
6	47	400	400	400	400	399	359	325
6.5	49	400	400	400	400	400	400	374
7	52	400	400	400	400	400	400	400
7.5	58	400	400	400	400	400	400	400

Slab Thickness (Inches)	11'-6	12'-0	12'-6	13'-0	13'-6	14'-0	14'-6	15'-0
5	217	198	181	166	152	140	129	119
5.5	255	232	212	194	178	164	151	140
6	295	269	245	225	207	190	176	162
6.5	339	309	282	259	238	219	203	188
7	384	350	320	294	270	249	230	213
7.5	400	390	357	327	301	278	257	237

16 ga Lightweight Concrete (115 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0	8'-6	9'-0	9'-6	10'-0	10'-6	11'-0
5	37	400	400	391	349	313	282	255
5.5	42	400	400	400	400	390	343	310
6	47	400	400	400	400	400	400	369
6.5	49	400	400	400	400	400	400	400
7	52	400	400	400	400	400	400	400
7.5	58	400	400	400	400	400	400	400

Slab Thickness (Inches)	11'-6	12'-0	12'-6	13'-0	13'-6	14'-0	14'-6	15'-0
5	231	211	193	177	162	149	138	127
5.5	262	237	215	195	178	162	148	136
6	303	275	250	227	206	187	171	156
6.5	343	314	287	262	240	221	204	188
7	400	400	378	347	320	298	273	253
7.5	400	400	400	391	361	333	308	286